Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 7
Hypothesis test

Hypothesis Testing of Normal Population Mean

◆U test

$$U = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1)$$

◆T test

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

Hypothesis testing steps

In general, hypothesis testing may be described as a five-step procedure:

- Step 1: H_0 and H_1. (Choose the value of α).
- Step 2: Conduct the test statistic.
- Step 3: Find out the rejection region
  (Reject H_0 |when H_0 is real） = α .
- Step 4: Calculate the test statistic of a sample.
- Step 5: Conclusion.

The p-value is the probability of obtaining a value for the test statistic that is more extreme than the value actually observed. (Probability is calculated under H_0 .)

Hypothesis Testing of Normal Population Mean

Example 3. Example 3: The magnetic intensity in a certain region is known to follow a normal distribution with parameters that μ_0=56 and σ_0 =20 based on previous observations. A new instrument is used to measure the magnetic intensity, and a sample of 41 measurements is taken. The sample mean and variance are $\bar{X}$=61.1, s=20. Does this instrument meet the requirements at the significance level α=0.05?

Hypothesis Testing of Normal Population Mean

Step 1: $H_0: \quad \mu = \mu_0 = 56 \qquad H_1: \quad \mu \neq \mu_0 = 56$

Step 2: $U = \dfrac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1) \qquad \alpha = 0.05$

Step 3: $P_{\mu_0}\{|U| \geq z_{\frac{0.05}{2}}\} = \alpha \qquad z_{\frac{0.05}{2}} = z_{0.025} = 1.96$

Rejection region: $|U| \geq 1.96$

Step 4 : $\left|\dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}\right| = \left|\dfrac{61.1 - 56}{20 / \sqrt{41}}\right| = \left|\dfrac{5.1}{3.125}\right| = 1.632 < 1.96$

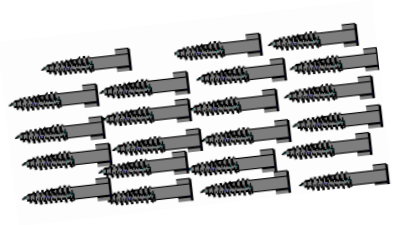
Step 5: H_0 cannot be rejected, H_0 should be accepted.

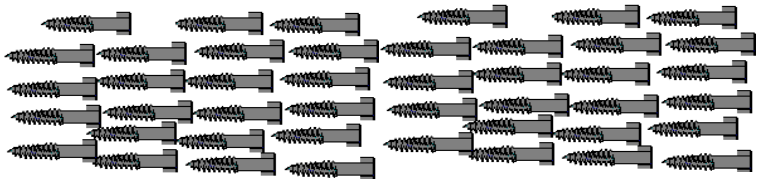
It can be considered that the machine is functioning properly.

Hypothesis Testing of Normal Population Mean

Example 3: The standard length of a screw produced by a factory is 32.5 mm. The actual length X of the produced screws is assumed to follow a normal distribution $N(\mu, \sigma^2)$, where σ^2 is unknown. A sample of 6 screws is randomly selected from a batch, and the following dimension data is recorded: 32.56, 29.66, 31.64, 30.00, 31.87, 31.03.

Is this batch of products acceptable?





Hypothesis Testing of Normal Population Mean

Step 1: $H_0 : \mu = 32.5 \quad \Leftrightarrow \quad H_1 : \mu \neq 32.5$

Step 2: $T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1) \quad \alpha = 0.01$

Step 3: $P\{|t| > t_{\alpha/2}(5)\} = \alpha \qquad t_{\alpha/2}(5) = t_{0.005}(5) = 4.0322$

Reject H_0: $C: \quad |t| > 4.0322$

Step 4:

$$t = \frac{\bar{X} - 32.5}{S/\sqrt{6}} \qquad |t| = 2.997 < 4.0322$$

Step 5: H_0 cannot be rejected, H_0 should be accepted.
Thus this batch of products can be considered as qualified.

P value

p-value Definition:

The p-value is $P(\text{data} \mid H_0)$, representing the probability of obtaining the observed data, or more extreme results, assuming the null hypothesis H_0 is true.

Interpretation of the p-value:

- Reject H_0 if $p \leq \alpha$
- Do not reject H_0 if $p > \alpha$

The p-value is valuable because it provides more nuanced information than a simple yes/no conclusion for the hypothesis test. The choice of significance level α is subjective and influences the decision rule. Two researchers examining the same data may reach different conclusions if they use different significance levels. For example, if researcher A prefers $\alpha = 0.01$ and researcher B prefers $\alpha = 0.05$, a p-value of 0.03 would lead to researcher A not rejecting H_0, while researcher B would reject H_0.

Hypothesis Testing of Normal Population Mean

- The alternative hypothesis H_1 suggests that μ may be either greater than or less than μ_0 , making it a two-tailed (bilateral) hypothesis test.
- **One-tailed hypothesis tests:**
  - $H_0 : \mu = \mu_0$
  - (1) Right-tailed test: $H_1 : \mu > \mu_0$
  - (2) Left-tailed test: $H_1 : \mu < \mu_0$